

KAVYA SREE CHANDHI

AI & Data Analytics Specialist | Business Intelligence • Python • SQL • Tableau • GenAI

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PROFESSIONAL SUMMARY:

AI & Data Analytics Specialist with 5+ years of experience building scalable data processing pipelines, interactive BI dashboards, and AI-driven automation systems. Proven domain background in **banking and financial services (BFSI)**, specializing in transaction risk analytics, anomaly detection, ETL automation, and regulatory compliance (KYC/AML). Proficient in **Python, SQL, Tableau, Grafana, and Generative AI frameworks** to transform complex datasets into actionable business intelligence and operational insights. Skilled in cross-functional collaboration with engineering, IT, and leadership teams to ensure high data integrity, strict security standards, and data-driven decision-making.

TECHNICAL SKILLS:

- **Programming & Core Computer Science:** Python, C, C++, JavaScript, SQL, Data Structures & Algorithms, Object-Oriented Programming (OOP), Memory Management, Multithreading, Operating Systems, System Design
- **Business Intelligence & Analytics:** Tableau, Power BI, Grafana, Dash, Plotly, Matplotlib, Microsoft Excel, Exploratory Data Analysis (EDA), KPI Tracking, Interactive Dashboard Design, Automated Reporting Workflows
- **Generative AI & LLM Engineering:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, Fine-Tuning, Multi-Agent Systems, LangChain, LangGraph, Hugging Face, OpenAI API, Ollama, DALL-E, Vector Databases (Milvus, ChromaDB, Pinecone)
- **Machine Learning & Data Science:** Supervised & Unsupervised Learning, Predictive Modeling, Feature Engineering, Cross-Validation, Hyperparameter Tuning, Statistical Analysis, Scikit-learn, Anomaly Detection
- **Deep Learning & Computer Vision:** Artificial Neural Networks (ANN), CNN, RNN, LSTM, ResNet, VGG, MobileNet, Transfer Learning, GANs, Autoencoders, PyTorch, TensorFlow, Keras, OpenCV
- **Natural Language Processing (NLP):** Sentiment Analysis, Text Summarization, Named Entity Recognition (NER), Emotion Detection, Text Classification, spaCy, NLTK
- **Recommender Systems & Optimization:** Collaborative Filtering, Matrix Factorization, Bayesian Optimization, Optuna, Hyperopt
- **Data Engineering, Big Data & Databases:** SQL Query Optimization, Complex Joins, MySQL, PostgreSQL, MongoDB, DynamoDB, Neo4j, Apache Spark, PySpark, Hadoop, Kafka, Spark Streaming, Snowflake, Redshift, Azure Data Lake, BigQuery
- **Cloud Platforms & Web Development:** AWS (SageMaker, Lambda, EC2, S3, RDS, Glue), Microsoft Azure (Prompt Flow, Bot Framework, Cosmos DB, Data Factory), RESTful APIs, FastAPI, Flask, Streamlit, Django
- **MLOps, DevOps & Software Engineering:** Docker, Kubernetes, Jenkins, GitHub Actions, MLflow, Weights & Biases (W&B), DVC, Terraform, Airflow, Helm, CI/CD Pipelines, STL, Pointers, File Handling, Debugging, Profiling, Performance Tuning
- **Governance, Compliance & Security:** Data Integrity, Data Quality Assurance, BFSI Regulations, KYC/AML Compliance, ISO/IEC Standards, HIPAA, GDPR, PCI DSS, Role-Based Access Control (RBAC)
- **Collaboration & Environment:** Git, GitHub, Jira, Confluence, Agile/Scrum, Jupyter Notebook, VS Code, PyCharm, Anaconda

WORK EXPERIENCE:

Client- UWorld,USA

Aug 2024 – Jan 2026

Role - Generative AI Engineer

Project Description: Leading the development of an AI-Enhanced Analytics & Test Prep Platform, implementing automated

data processing workflows, performance telemetry monitoring, evaluation pipelines, and real-time user engagement tracking.

Key Responsibilities:

- Built tracking pipelines using **Azure Application Insights** to monitor real-time user activity, system uptime, and feature accuracy.
- Optimized database queries and response times to help students access study materials and test data without delay.
- Created automated data workflows using **Azure OpenAI** and **LangGraph** to reduce manual effort in content review and testing.
- Built search and retrieval pipelines using **ChromaDB** so users could easily query large sets of study and analytical data.
- Developed **automated evaluation systems** using standardized rubrics to deliver fast feedback and performance reports.
- Designed user activity tracking tools and leaderboards that helped increase daily student participation.
- Partnered directly with **product managers and engineers** to turn platform data into clear updates for team decision-making.

Client – DXC Technology, India

Aug 2022 – Aug 2023

Role – Associate Professional Software Engineer

Project Description: Enhanced engineering workflows for automotive clients using Python automation, SQL reporting pipelines, and interactive data dashboards.

Key Responsibilities:

- Built **Python automation tools** to handle repetitive data tasks, saving time for engineering teams.
- Wrote and optimized **SQL queries** to speed up data extraction and support business reporting.
- Designed interactive dashboards in **Tableau** and **Grafana** to monitor software license usage and system performance.
- Managed user accounts and access controls for over **5,000 enterprise users**, ensuring data security.
- Integrated **predictive maintenance models** to catch system issues early and prevent downtime.
- Set up **CI/CD deployment pipelines** using Docker and Jenkins to release script updates faster.
- Followed strict **ISO/IEC security standards** to keep company data safe and compliant.
- Wrote technical guides and standard operating procedures (SOPs) to help onboard new team members.

Client – Pelican IT Solutions Pvt. Ltd., India

Feb 2020 – July 2022

Role – Data Analyst

Project Description: Supported a leading banking client in developing a **Fraud Detection & Risk Analytics Platform** to identify high-risk financial transactions, reduce fraud losses, and maintain regulatory compliance.

Key Responsibilities:

- Developed **Python and SQL data processing pipelines** to analyze large-scale, high-volume financial transaction data.
- Built **statistical and machine learning models** (such as Random Forest, Isolation Forest, and Autoencoders) to detect transaction anomalies, improving fraud detection accuracy by **20%**.
- Automated **ETL workflows** using **Python (Pandas, NumPy, Scikit-learn)** to clean and standardize financial records, reducing manual processing effort by **30%**.
- Designed real-time fraud monitoring dashboards using **Flask and Grafana** to provide stakeholders with immediate visibility into high-risk account activities.
- Created key fraud risk indicators, including **transaction velocity, user behavioral patterns, and seasonal trends**, to strengthen anomaly reporting.

- Applied predictive analytics and parameter tuning techniques to ensure robust, data-driven fraud risk assessments.
- Collaborated directly with **compliance and risk teams** to ensure all analytics and reporting aligned strictly with **BFSI regulations (KYC/AML)**.
- Streamlined real-time data pipelines with data engineering teams to accelerate transaction scoring speed across fraud detection workflows.

VOLUNTEERING:

Client – Network Theory Applied Research Institute

Jun 2025 – Aug 2025

Role – Software Engineer (Volunteer)

Project Description: Contributed to the development and enhancement of NTARI's internal research operations platform, focusing on frontend development and backend system analysis within an Agile team environment.

Key Responsibilities:

- Developed and enhanced frontend user interface components to improve usability, responsiveness, and overall user experience.
- Contributed to frontend architecture implementation, ensuring modular and maintainable code structure.
- Conducted detailed backend file and data structure analysis to understand system logic and integration points.
- Collaborated with cross-functional teams in an Agile environment to deliver incremental feature updates.
- Assisted in aligning frontend components with backend workflows to ensure accurate data representation and system consistency.
- Participated in sprint discussions, issue tracking, and iterative development cycles to support timely feature releases.
- Identified opportunities for performance improvements and workflow optimization across system components.

Client –Street Cause SWEC (NGO Initiative)

July 2020 – July 2021

Role – Associate President – Executive Board (2020–2021)

Project Description: Led social impact initiatives focused on community welfare during the COVID-19 pandemic, coordinating large-scale relief drives including slum adoption programs and public health awareness campaigns. Managed cross-functional volunteer teams to execute essential supply distribution, mask drives, and community outreach programs impacting underserved populations.

Key Responsibilities:

- Co-led strategic planning and execution of Slum Adoption Project, supporting 10+ slums and 262+ families with monthly essential groceries.
- Organized and managed COVID-19 Mask Distribution Drive (June 2020), ensuring community safety awareness and protective supply distribution.
- Supervised multiple functional departments including Women Empowerment, Education, and Orphanage & Old Age Homes initiatives.
- Coordinated volunteer teams, task delegation, and execution timelines to ensure smooth program delivery.
- Managed stakeholder communication, partnerships, and operational planning.
- Contributed to digital campaign promotion and awareness initiatives through social media outreach.
- Strengthened organizational governance and operational efficiency as part of the Executive Board.

ACADEMIC PROJECTS:

Youth Smoking and Drug Use Analysis

- Built data visualizations to present trends and key insights, showing that peer influence and school-based programs were major factors in usage patterns. Conducted exploratory data analysis (EDA) and statistical modeling on 8,900+ survey records using Python to identify key social factors influencing substance use.

AI Research Agent for Healthcare Diagnostics

- Developed an AI research assistant using Python, Streamlit, LangChain, and LLaMA3 with a RAG pipeline to retrieve and analyze medical data. Implemented automated citation checks to ensure high data accuracy and generate clear, structured diagnostic reports.

Automatic Attendance Monitoring & Analytics System

- Created an automated attendance system using Python, OpenCV, and Django with integrated machine learning classifiers. Designed interactive dashboards and role-based tracking tools to eliminate proxy attendance and streamline administrative reporting workflows.

Securing Data Using Blockchain and AI

- Built a secure prototype architecture using Django, Blockchain, and AI anomaly detection to protect sensitive record workflows. Enabled patient-controlled data access controls while ensuring tamper-proof record management and data integrity.

CERTIFICATIONS:

- **Microsoft Azure Fundamentals (AZ - 900)** - [Microsoft](#)
- **Artificial Intelligence Fundamentals** - [IBM](#)
- **Google Technical Support Fundamentals** – [Coursera/ Google](#)

ACADEMICS:

Master of Science in Computer Science | GPA: 3.916/4

Aug 2023 – May 2025

University of North Texas, Denton, Texas

- **Relevant Coursework:** Machine Learning, Artificial Intelligence, Big Data Analytics, Advanced Database Systems, Cloud Computing, Software Architecture
- **Key Focus Areas:** Predictive Modeling, Retrieval-Augmented Generation (RAG), ETL Pipelines, Anomaly Detection, Enterprise System Design